

# CSE 330 Assignment-4

**Deadline: 04 January 2026**

## Question 1

A linear system is described by the following equations:

$$x_1 + 6x_2 + 2x_3 = 10,$$

$$3x_1 + 2x_2 + x_3 = 6,$$

$$4x_1 + 5x_2 + 2x_3 = 9.$$

Based on these equations, answer the questions below.

- (a) From the given linear equations, identify the matrices  $A$ ,  $\mathbf{x}$ , and  $\mathbf{b}$  such that the linear system can be expressed in the matrix form

$$A\mathbf{x} = \mathbf{b}.$$

- (b) Construct the Frobenius matrices  $F^{(1)}$  and  $F^{(2)}$  corresponding to this system.
- (c) Compute the unit lower triangular matrix  $L$ .
- (d) Using the LU decomposition method, find the solution of the linear system. Use the unit lower triangular matrix obtained in part (c).

## Question 2

A linear system is described by the following equations:

$$6x_2 + 2x_3 = 10,$$

$$3x_1 + 2x_2 + x_3 = 6,$$

$$4x_1 + 5x_2 + 2x_3 = 9.$$

Based on these equations, answer the questions below.

- (a) From the given linear equations, identify the matrices  $A$ ,  $\mathbf{x}$ , and  $\mathbf{b}$  such that the linear system can be written in matrix form.

- (b) Examine whether the matrix  $A$  has any pivoting problem. Clearly explain your reasoning.
- (c) Write down the augmented matrix  $\text{Aug}(A)$  for the given linear system and evaluate the upper triangular matrix  $U$ . Show the row multipliers  $m_{ij}$  for each elimination step where necessary.
- (d) Using the upper triangular matrix obtained in part (c), compute the solution of the given linear system using the Gaussian elimination method.